

Ch-1:-

Sets.

Defⁿ:- A Set is a well defined collection of objects.

Note:- (i) Sets are usually denoted by Capital Letters (A, B, C, ...)
(ii) The elements of sets are represented by small letters (a, b, c, ...).

N - Natural, W - Whole, I/Z - Integer, Q - Rational, Q^c - Irrational
R - Real No.

→ If (small) "a" is an element of Set "A", then we say

"a belongs to A"
We write it as " $a \in A$ "

→ If "a" is not an element of Set "A", then we say

"a does not belong to A" → " $a \notin A$ "

Representation of a Set:-

I) Roster or Tabular form:- In it, all the elements of sets being separated by "," & enclosed with "{ }".

eg:- a) Set of all N < 5 ⇒ {1, 2, 3, 4}
b) " " all Vowels ⇒ {a, e, i, o, u}

II) Set Builder form: In it, all the elements of a set possess a single common property which is not possessed by any element outside the set.

eg: a) $\{x | x \text{ is a } N < 5\}$ b) $\{x | x \text{ is vowel in Eng. Alph}\}$

Q. Write the following Set in Roster form:

i) $A = \{x | x \text{ is an integer \& } -3 < x < 7\}$

Sol: $\{-2, -1, 0, 1, 2, 3, 4, 5, 6\}$

ii) $B = \{x | x \text{ is 2-digit } N, \text{ Sum} = 8\}$

Sol: $\{17, 26, 35, 44, 53, 62, 71, 80\}$

iii) $C = \{x | x \text{ is letters in word 'BETTER'}\}$

Sol: $\{B, E, T, R\}$

Note:- The Roster form of a set is unique, but Set builder is not unique.

eg: $\{1, 2\} \rightarrow A = \{x | x \text{ is } N < 3\}$

$B = \{x | x \text{ is the sol}^n \text{ of eqn } (x-1)(x-2)=0\}$

eg: $\{3, 6, 9, 12\} \rightarrow A = \{x | x \text{ is multiple of 3, } < 15\}$

$B = \{x | x = 3n, n \in N < 5\}$

→ Finite Set

Empty Set:- A Set which does not contain any element.

→ Also known as:- Null Set, Void Set.

→ We represent it by $\emptyset$ or $\{\}$ ← empty brackets.

eg:- $\{x | x \in \mathbb{N} \text{ \& } 1 < x < 2\} \rightarrow \{\}$

Finite & Infinite Set:-

Finite:- A Set which is Empty or contain finite no. of element.

Infinite:- " " " don't contain finite no. of element.

eg:- F :- ω be the Set of days of week.

I :- G be the Set of points on the line.

Q. Which are F or I Set:-

i) $\{x | x \in \mathbb{N} \text{ \& } (x-1)(x-2) = 0\} \Rightarrow \text{Finite}$

ii) $\{x | x \in \mathbb{N} \text{ \& } x^2 = 4\} \Rightarrow \text{Finite}$

iii) $\{x | x \in \mathbb{N} \text{ \& } x \text{ is prime}\} \Rightarrow \text{Infinite}$

iv) $\{x | x \in \mathbb{N} \text{ \& } x \text{ is odd}\} \Rightarrow \text{Infinite}$

Equal Set:- Two Sets A & B are said to be equal, if they have exactly same element. And we write $A = B$.

→ Otherwise, the Sets are said to be unequal, write it as $A \neq B$.

eg:- $A = \{1, 2, 3, 4\}$; $C = \{a, b, c, d\}$ } $A = B$; $B = D$; $A = D$
 $B = \{4, 3, 1, 2\}$; $D = \{2, 1, 4, 3\}$ } but $\Rightarrow C \neq A$; $C \neq B$; $C \neq D$.

Quadratic Eqⁿ:- $\frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

Q. Find pair of equal sets:-

i) $A = \{0\} \Rightarrow \{0\}$
 $B = \{x | x > 15 \& x < 5\} \Rightarrow \{\} \text{ or } \emptyset$
 $C = \{x | x - 5 = 0\} \Rightarrow \{5\}$
 $D = \{x | x^2 = 25\} \Rightarrow \{-5, 5\}$
 $E = \{x | x \text{ is +ve root of eq}^n \Rightarrow \{+5\}$
 $x = +5 \text{ or } -3 \quad x^2 - 2x - 15 = 0 \} \Rightarrow$

ii) A, the set of letters in word 'ALLOY' $\Rightarrow \{A, L, O, Y\}$
 B, " " " " " " 'LOYAL' $\Rightarrow \{L, O, Y, A\}$ equal.

iii) $A = \{n | n \in \mathbb{Z} \& n^2 \leq 4\} \Rightarrow \{-2, -1, 0, 1, 2\}$
 $B = \{x | x \in \mathbb{R} \& x^2 - 3x + 2 \geq 0\} \Rightarrow \{2, 1\}$ not equal.

$0 \in A$ but $0 \notin B \Rightarrow A \neq B$.

Subsets:- A Set 'A' is said to be subset of Set "B", if every element of A is also an element of B.

Symbol
Represented by:- $A \subset B / A \subsetneq B$

eg:- $A = \{1, 2, 3, 4\}$; $B = \{1, 2, 4\} \Rightarrow B \subset A$
 $\rightarrow B \subset A$; if whenever $a \in A$, and $a \in B$

$\rightarrow$ empty set is subset of every set.

$\rightarrow$ In equal set if $A = B \Rightarrow A \subset B$ and $B \subset A$.

$\rightarrow$ Every set is subset of itself.

Q. $\phi, A = \{1, 3\}, B = \{1, 5, 7\}, C = \{1, 3, 5, 7, 9\}$

Insert $\subset$ & $\not\subset$

a) $\phi \subset B$

b) $A \not\subset B$

c) $A \subset C$

d) $B \subset C$

Subsets of Real No.:-

$N = \{1, 2, 3, 4, \dots\}$

$W = \{0, 1, 2, 3, \dots\}$

$Z = \{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}$

$Q = \{x | x = \frac{p}{q} \text{ \& } q \neq 0\}$

$N \subset W$

$W \subset Z$

$Z \subset Q$

$Q \subset R$

$Q \subset R$

$N \subset W \subset Z \subset Q \subset R$

Interval as Subsets of real no.:-

a) $(1, 3) = \{x | 1 < x < 3\}$; c) $[1, 3) = \{x | 1 \leq x < 3\}$

b) $[1, 3] = \{x | 1 \leq x \leq 3\}$; d) $(1, 3] = \{x | 1 < x \leq 3\}$

→ Let a, b belong to Real no. and $a < b$,

→ When, the set of real no. is $\{x | a < x < b\}$, called open interval, Denoted as (a, b) .

→ " " " " " " " " $\{x | a \leq x \leq b\}$, close interval, " " $[a, b]$.

→ Semi-Open & Semi-Close $\Rightarrow (a, b], [a, b)$.

Power-Set:- The Set of all possible subsets of a Set 'A' is called, Power-Set.

Denoted by:- $P(A)$.

eg:- $A = \{1, 2, 3\} \Rightarrow \phi, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{2, 3\}, \{1, 3\}, \{1, 2, 3\}$
Sub sets.

$\{\phi, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{2, 3\}, \{1, 3\}, \{1, 2, 3\}\}$
Power set

eg. $B = \{1, 4\} \Rightarrow \underbrace{\{\emptyset, \{1\}, \{4\}, \{1, 4\}\}}_{\text{Set}} \leftarrow \text{Power set } P(B)$

Cardinality of a Set :- the no. of element in a set A is called Cardinality of that Set A .

Denoted by:- $n(A)$

eg. $A = \{a, b, c, d, e\} \Rightarrow n(A) = 5$

• Note:- Let A be a set with $n(A) = m$, then $n(P(A))$ is 2^m . (See example of Power set).

Universal Set:- Denoted by (usually) $= U$
A set which contain all elements of given Subsets, also called Basic-Set.

Q. Write all subset of following Sets:-
Power set.

i) $\{a\} \Rightarrow \{\emptyset, \{a\}\}$

ii) $\emptyset \Rightarrow \{\emptyset\}$
 $n(P(\emptyset)) = 1$
 $n(\emptyset) = 0$
 $P(\emptyset) = 1$

iii) $\{1, 2, 3, 4\} \Rightarrow \{\emptyset, \{1\}, \{2\}, \{3\}, \{4\}, \{1, 2\}, \{2, 3\}, \{3, 4\}, \{1, 4\}, \{1, 2, 3\}, \{2, 3, 4\}, \{1, 3, 4\}, \{1, 2, 4\}, \{1, 3, 4\}, \{2, 4\}, \{1, 2, 3, 4\}\}$

iv) $\{a, b\} = \{\emptyset, \{a\}, \{b\}, \{a, b\}\}$

Q. Write the following as Interval:-

1) $\{x | x \in \mathbb{R}, -4 \leq x \leq 6\} \Rightarrow [-4, 6]$

2) $\{x | x \in \mathbb{R}, -12 \leq x < -10\} \Rightarrow [-12, -10)$

3) $\{x | x \in \mathbb{R}, 0 \leq x < 7\} \Rightarrow [0, 7)$

4) $\{x | x \in \mathbb{R}, 3 \leq x \leq 4\} \Rightarrow [3, 4]$

Q. Write following in Set-builder form:-

- 1) $(-3, 0) \Rightarrow \{x \mid -3 < x < 0\}$
 2) $[6, 12] \Rightarrow \{x \mid 6 \leq x \leq 12\}$
 3) $(6, 12] \Rightarrow \{x \mid 6 < x \leq 12\}$
 4) $[-23, 5) \Rightarrow \{x \mid -23 \leq x < 5\}$

Q. Given $\Rightarrow A = \{1, 3, 5\}$ $B = \{2, 4, 6\}$ $C = \{0, 2, 4, 6, 8\}$

$U = ?$

- a) $\{0, 1, 2, 3, 4, 5, 6\}$ b) $\emptyset$
 ✓ $\{0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$ d) $\{1, 2, 3, 4, 5, 6, 7, 8\}$

Operation on Set:-

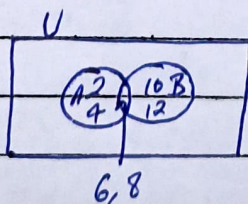
1) Union of Sets:- Let, A & B any 2 sets, the union of A & B is the set which contains all the elements of A & B.

$\rightarrow$ The common elements been taken only once

$\rightarrow$ Symbol:- "U" & written as $A \cup B$

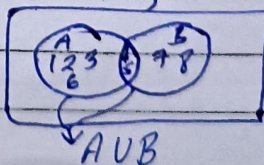
Q. if $A = \{2, 4, 6, 8\}$ & $B = \{6, 8, 10, 12\}$; find $A \cup B$ and make Venn Diagram.

$A \cup B = \{2, 4, 6, 8, 10, 12\}$



eg:- $A = \{1, 2, 3, 4, 5, 6\}$; $B = \{4, 5, 7, 8\}$

$A \cup B = \{1, 2, 3, 4, 5, 6, 7, 8\}$



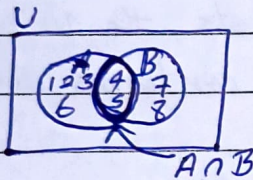
→ Properties of Union:-

- i) Commutative $\Rightarrow A \cup B = B \cup A$
- ii) Associative $\Rightarrow (A \cup B) \cup C = A \cup (B \cup C)$
- iii) $A \cup \emptyset \Rightarrow A$
- iv) $A \cup U \Rightarrow U$
- v) $A \cup A \Rightarrow A$

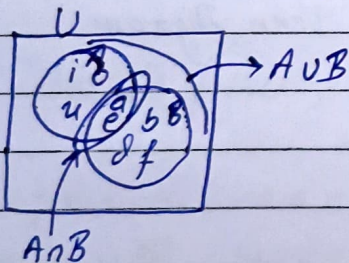
2) Intersection of Sets:- The intersection of set A & B is the set of all element which are common in both A & B.

→ Symbol is " $\cap$ ", written $\Rightarrow A \cap B$.

eg:- $A = \{1, 2, 3, 4, 5, 6\}$; $B = \{4, 5, 7, 8\}$
 $A \cap B = \{4, 5\}$



d. $A = \{a, e, i, o, u\}$ & $B = \{b, a, e, d, f\}$
 find $A \cup B = \{a, e, i, o, u, b, d, f\}$
 $A \cap B = \{a, e\}$



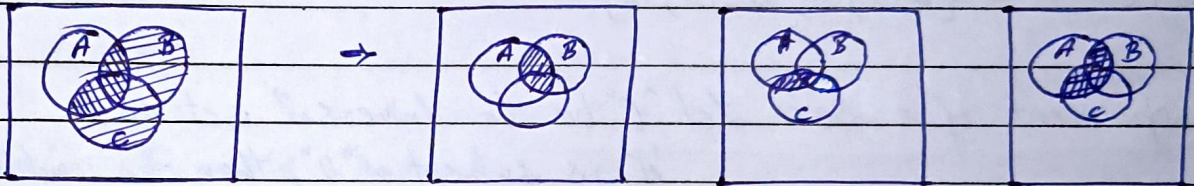
→ Properties of Intersection:-

- i) Commutative $\Rightarrow A \cap B = B \cap A$
- ii) Associative $\Rightarrow (A \cap B) \cap C = A \cap (B \cap C)$
- iii) $A \cap \phi = \phi$
- iv) $A \cap U = A$
- v) $A \cap A = A$

3) Distributive Law:-

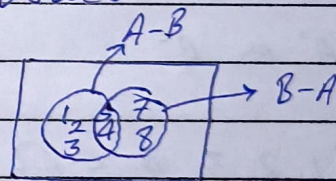
$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$



* **Difference of Set**:- The diff. of the sets A and B in this order is the set of order.

$$A - B = \{x / x \in A \text{ \& } x \notin B\}$$



$$A = \{1, 2, 3, 4, 5, 6\} \quad ; \quad B = \{4, 5, 6, 7, 8\}$$

$$A - B = \{1, 2, 3\} \quad ; \quad B - A = \{7, 8\}$$

d. If $A = \{3, 6, 9, 12, 15, 18, 21\}$

$$B = \{4, 8, 12, 16, 20\}$$

$$C = \{4, 6, 8, 10, 12, 14, 16\}$$

$$D = \{5, 10, 15, 20\}$$

find:- i) $A \cup B \Rightarrow \{3, 6, 9, 12, 15, 18, 21, 4, 8, 16, 20\}$

ii) $A \cap C \Rightarrow \{6, 12\}$

iii) $C \cap D \Rightarrow \{10\}$

iv) $A \cap D \Rightarrow \{15\}$

v) $A - B \Rightarrow \{3, 6, 9, 15, 18, 21\}$

vi) $B - A \Rightarrow \{4, 8, 16, 20\}$

vii) $B - C \Rightarrow \{20\}$

viii) $B - D \Rightarrow \{4, 8, 12, 16\}$

ix) $D - B \Rightarrow \{5, 10, 15\}$

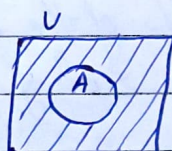
x) $D - C \Rightarrow \{5, 15, 20\}$

xi) $A - C \Rightarrow \{3, 9, 15, 18, 21\}$

xii) $A - D \Rightarrow \{3, 6, 9, 12, 18, 21\}$

xiii) $C - D \Rightarrow \{4, 6, 8, 12, 14, 16\}$

5) Complement of a Set:- Let U be the Universal set and A is subset of U , then the complement of A is the set of all element of U which are not the element of A .



Write A' or A^c

eg:- $U = \{1, 2, 3, 4, 5, 6\}$

$A = \{5, 6\}$

$A^c = \{1, 2, 3, 4\}$

Q.1- $U = \{1, 2, 3, 4, 5, 6\}$

$A = \{2, 3\}$; $B = \{3, 4, 5\}$

find A' & B'

$(A \cup B)' = A' \cap B'$

$A' = \{1, 4, 5, 6\}$

$B' = \{1, 2, 6\}$

$(A \cup B)' = \{1, 6\}$

$(A \cap B)' = \{1, 6\}$

$A \cup B = \{3, 2, 4, 5\}$

$A' \cap B' = \{1, 6\}$

Complement Properties:-

i) Complement Law:-

$$a) A \cup A' = U$$

$$b) A \cap A' = \phi$$

ii) Law of Double Complements:-

$$(A')' = A$$

iii) Law of empty set & Universal set:-

$$\phi = U$$

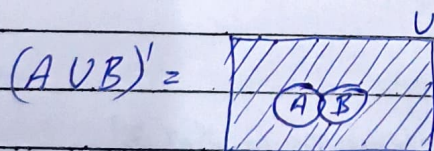
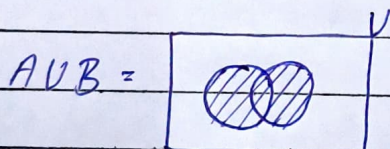
$$U' = \phi$$

iv) De Morgan's Law:-

$$a) A \cup B' = A' \cap B'$$

LHS:- $(A \cup B)'$:-

RHS:-



Q. $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$; $A = \{1, 2, 3, 4\}$
 $C = \{3, 4, 5, 6\}$; $B = \{2, 4, 6, 8\}$

find- i) $A' = \{5, 6, 7, 8, 9\}$

ii) $B' = \{1, 3, 5, 7, 9\}$

iii) $(A \cup C)' = \{7, 8, 9\}$

iv) $(A \cup B)' = \{5, 7, 9\}$

v) $(A')' = \{1, 2, 3, 4\}$

vi) $(B - C)' = \{1, 4, 6, 7\}$

$\Rightarrow \{2, 8\} \Rightarrow \{1, 3, 4, 5, 6, 7, 9\}$

Q. a) $A \cup A' = U$

c) $A \cap A' = \emptyset$

b) $\emptyset' \cap A = A'$

d) $U' \cap A = \emptyset$

6) Cartesian Product of sets:- Given 2 non empty Set P & Q , the Cartesian Product $P \times Q$ is the Set of all $\downarrow$ img. Set of P order pairs of element.

$P = \{a, b\}$, $Q = \{1, 2, 3\}$

$P \times Q = \{(a, 1), (a, 2), (a, 3), (b, 1), (b, 2), (b, 3)\}$

$Q \times P = \{(1, a), (1, b), (2, a), (2, b), (3, a), (3, b)\}$

Note- 1) $(x, y) = (3, 4)$ $\xRightarrow{\text{then}}$ $x = 3$ & $y = 4$

2) Don't follow commutative law.

3) if there are 'n' element in Set 'A' & 'm' elements in Set 'B', then Cardinality of $A \times B \Rightarrow \underbrace{n \times m}_{\text{cross}} \Rightarrow n(A) = n \xrightarrow{\text{mak}} n(A \times B) = n \times m$
 $n(B) = m$

4) if A & B are non-empty sets and either A or B is an ∞ set then $A \times B = \{\infty\}$.

5) if one of A or B is an empty set ($\emptyset$), then $A \times B = \emptyset$.

Q. if $(x+1, y-2) = (3, 1)$; find x, y :-

$$x+1=3$$

$$\boxed{x=2}$$

$$y-2=1$$

$$\boxed{y=3}$$

Q. if $P = \{1, 2\}$; $P \times P = \{(1, 1), (1, 2), (2, 1), (2, 2)\}$

$$P \times P \times P = \{(1, 1, 1), (1, 1, 2), (1, 2, 1), (1, 2, 2), (2, 1, 1), (2, 1, 2), (2, 2, 1), (2, 2, 2)\}$$

Q. if $A = \{c, d\}$; $B = \{1, 2\}$

$$A \times B \times A = \{(c, 1, c), (c, 1, d), (c, 2, c), (c, 2, d), (d, 1, c), (d, 1, d), (d, 2, c), (d, 2, d)\}$$

Q. if $A = \{1, 2, 3\}$; $B = \{3, 4\}$; $C = \{4, 5, 6\}$

$$B \cup C = \{3, 4, 5, 6\}$$

$$i) A \times (B \cap C) = \{1, \cancel{2}, \{2, \cancel{4}\}, \{3, \cancel{4}\}$$

$$iii) A \times (B \cup C) = \{(1, 3), (1, 4), (1, 5), (1, 6), (2, 3), (2, 4), (2, 5), (2, 6), (3, 3), (3, 4), (3, 5), (3, 6)\}$$

$$ii) (A \times B) \cap (A \times C) \Rightarrow \{(1, 3), (1, 4), (2, 3), (2, 4), (3, 3), (3, 4) \cap (1, 4), (1, 5), (1, 6), (2, 4), (2, 5), (2, 6), (3, 4), (3, 5), (3, 6)\} \\ \Rightarrow \{(1, 4), (2, 4), (3, 4)\}$$

$$iv) (A \times B) \cup (A \times C) \Rightarrow \{(1, 3), (1, 4), (2, 3), (2, 4), (3, 3), (3, 4), (1, 5), (1, 6), (2, 5), (2, 6), (3, 5), (3, 6)\}$$

→ Relation:- Relation " R " from non-empty " A " to non-empty set " B " is a subset of cartesian product $A \times B$.

→ The subset is derived by describing a relationship b/w the first & second element of ordered pair $A \times B$.

→ The second element is called image of first element.

→ Domain:- the set of ^{all} first element of the ordered pair in a relation are from a set " A " to a set " B " is called the domain of Relation " R ".

→ Co-Domain & Range:- The set of all second element in a relation are from set " A " to set " B " is called range of Relation " R ".

→ The whole set " B " is called Co-domain of Relation " R ".

Note:- $\text{Range} \subseteq \text{Co-domain}$.

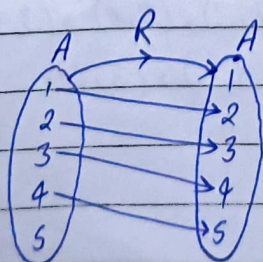
⇒ Representation of Relation

$$A = \{1, 2, 3, 4\} ; R: A \rightarrow A$$

i) Roster form:- $R = \{(1, 2), (2, 3), (3, 4), (4, 5)\}$

ii) Set Builder form:- $\{(x, y) \mid y = x + 1 \ \& \ x \in A, \text{ where } x < 5\}$
or where $x = 1, 2, 3, 4$
or $x = N ; x < 5$

iii) Line Diagram:-

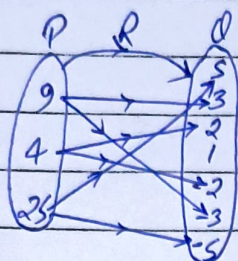


$$\text{Domain} = \{1, 2, 3, 4\}$$

$$\text{Co-Domain} = A = \{1, 2, 3, 4, 5\}$$

$$\text{Range} = \{2, 3, 4, 5\}$$

d. Write roster, Set-builder, domain, Co-domain, range.



Roster :- $R = P \rightarrow Q \Rightarrow \{(9, 5), (9, 3), (9, 2), (9, 1), (9, -2), (9, -3), (9, -5), (4, 5), (4, 3), (4, 2), (4, 1), (4, -2), (4, -3), (4, -5), (25, 5), (25, 3), (25, 2), (25, 1), (25, -2), (25, -3), (25, -5)\}$

Set-Builder :- $\{(x, y) \mid x = y^2 \text{ \& } y \in Q, \text{ where } y \neq 1\}$
 $y = \sqrt{x} \text{ \& } x \in P$

Domain :- $\{9, 4, 25\}$

Co-Domain :- $\{-5, -3, -2, 1, 2, 3, 5\}$

Range :- $\{5, 3, 2, -2, -3, -5\}$